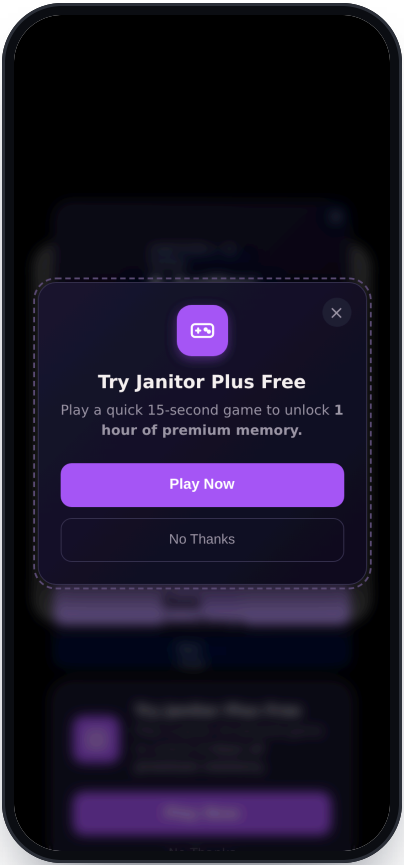
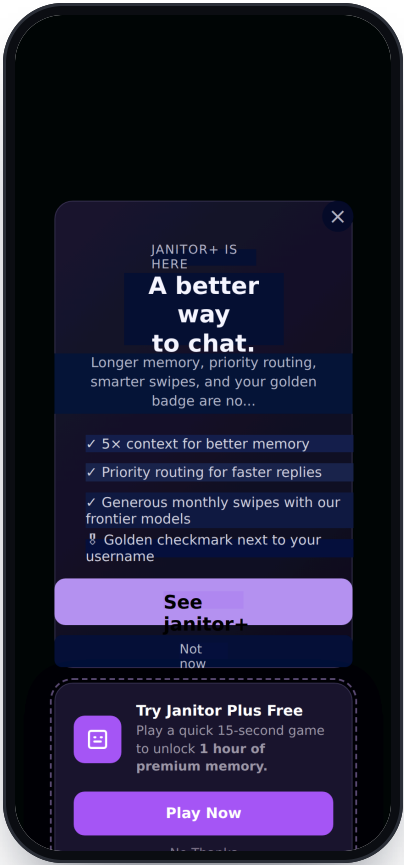


Let users **play a short game for a reward.**

Add two opt-in rewarded flows to Janitor that trade a short sponsored game for something users already value. We reviewed 3 ideas: 2 ship, 0 need revision, 1 rejected.



1 FLOW 1 · SLIDE 3

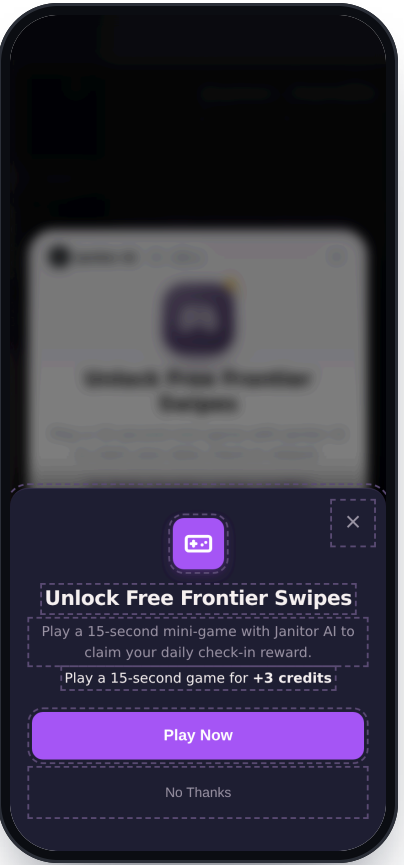
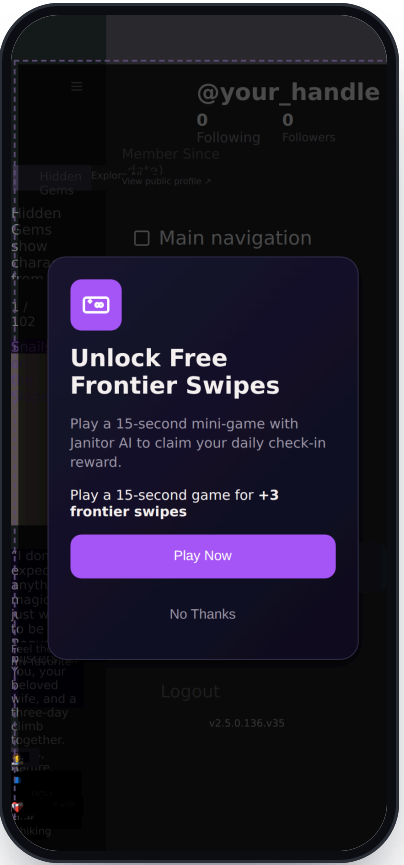
Try Janitor Plus Free for one short game

Offer a 1-hour trial of Janitor Plus when users dismiss the paywall after a rewarded mini-game.

SHIP

Score 4.90 / 5 (ship ≥ 3.8)

Product change



2 FLOW 2 · SLIDE 5

Profile menu: play for Daily frontier swipe credits

Earn daily check-in rewards and frontier model swipes by completing character quests via rewarded mini-games.

SHIP

Score 4.25 / 5 (ship ≥ 3.8)

Product change

EXCHANGE RATE

Not computable

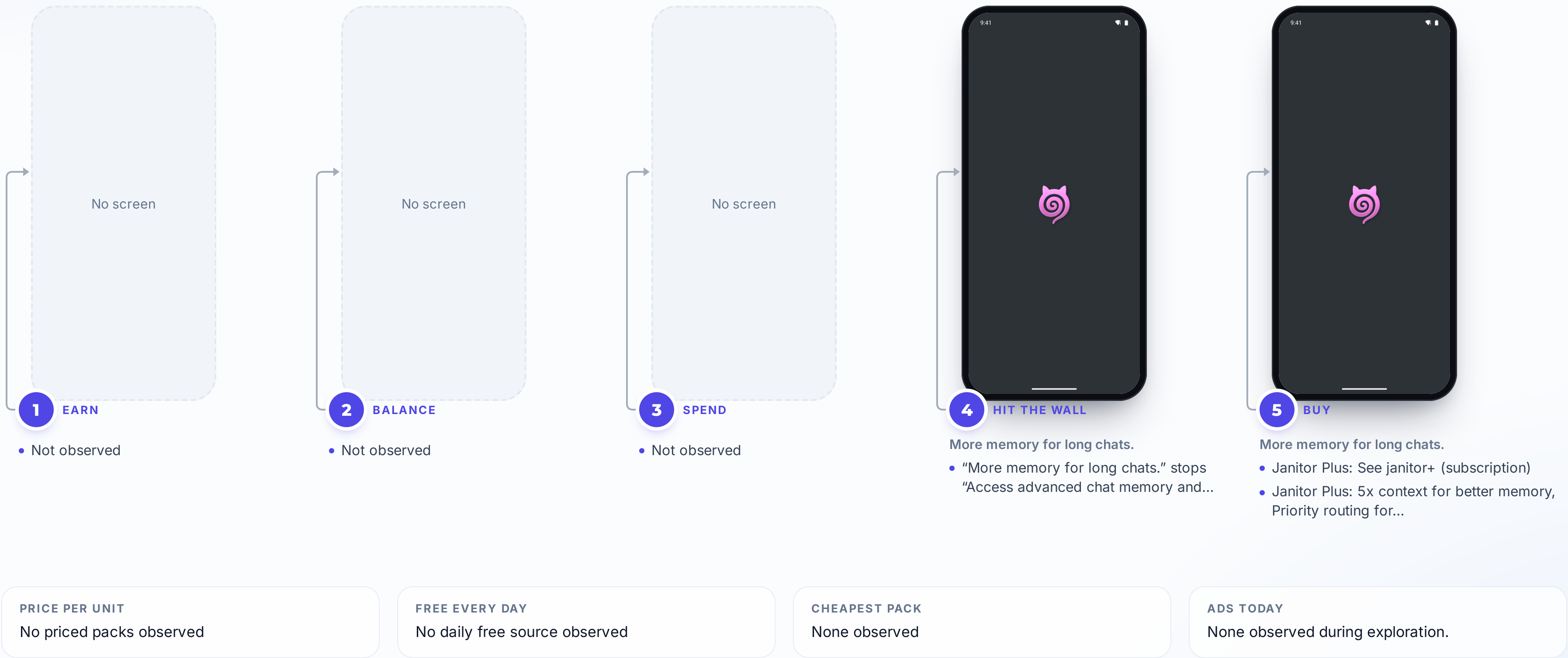
No priced packs observed: unit prices, action costs and exchange rate are unavailable.

CHEAPEST PACK

No priced pack was observed, so there is no paid path to cannibalize in currency terms.

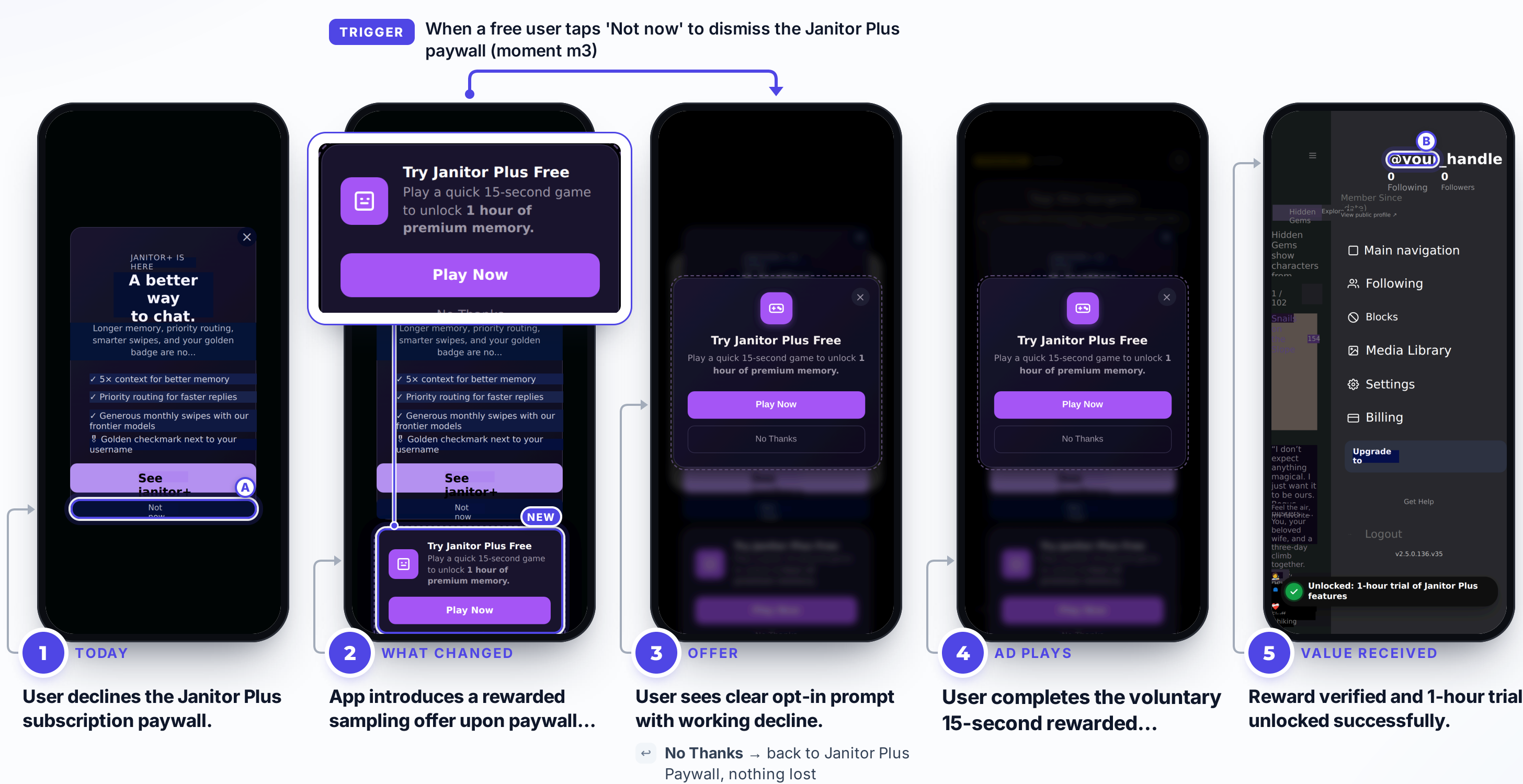
How Janitor makes money today

Through the Janitor Plus subscription tier, which unlocks enhanced memory context, priority routing, monthly model swipes, and a golden username badge. Real screens from the app; prices and costs as observed.



Try Janitor Plus Free for one short game

Offer a 1-hour trial of Janitor Plus when users dismiss the paywall after a rewarded mini-game.



SHIP Score 4.90 / 5 (ship ≥ 3.8)

Product change

WHY THIS WORKS

\$0.009–\$0.015 per view

Earned per completed US view; the reward costs ≈ \$0.0018 to serve.

Paid path untouched

Gated strictly to users who decline the paywall; time-boxed to 1 hour; permanently distinct from full monthly...

≤ 1 a day

Opt-in only; nothing plays unless the user taps.

Open the clickable prototype →

Simula SIM-RWD · invitation entry · Game Partner: Janitor · min play 15s · reward granted on REWARD_VERIFIED · frame callouts on slide 4

Subscription Feature Sampling via Rewarded Ad

TRIGGER

When a free user taps 'Not now' to dismiss the Janitor Plus paywall (moment m3).

ELIGIBILITY

Non-paying free users who dismiss the subscription paywall, capped at once per day.

WHAT THE USER SEES

Try Janitor Plus Free Play a quick 15-second game to unlock 1 hour of premium memory.

Play Now

No Thanks

REWARD AND CAPS

1-hour trial of Janitor Plus features, 1 hour; granted on REWARD_VERIFIED. At most 1 a day.

FRAME BY FRAME (SLIDE 3)

- 1

Today

User declines the Janitor Plus subscription paywall.

A

User taps 'Not now' on paywall
- 2

What changed

App introduces a rewarded sampling offer upon paywall decline.

NEW

trial offer button appears
- 3

Offer

User sees clear opt-in prompt with working decline.

Play Now: starts a 15s sponsored game

No Thanks: dismisses the offer

Clear disclosure of 1-hour trial
- 4

Ad plays

User completes the voluntary 15-second rewarded mini-game.

15-second partner mini-game
- 5

Value received

Reward verified and 1-hour trial unlocked successfully.

B

1-hour trial active

ECONOMICS (COMPUTED IN CODE)

Revenue per completed US view	\$0.009–\$0.015
Reward value at list price	n/a (no priced unit)
Reward ÷ one view	n/a
Max earnable per day (1/day)	n/a
Cheapest pack	none observed
Cost to serve the reward	\$0.0018 per view

SCENARIO, NOT FORECAST

0.25

\$0.003

\$1,095,000

views per DAU

ARPDau

per year per 1M DAU

Scenario, not a forecast: 25% of DAU engage, 1 views each, US mid eCPM after a 25% non-game haircut.

CANNIBALIZATION GUARD

Gated strictly to users who decline the paywall; time-boxed to 1 hour; permanently distinct from full monthly subscription entitlement.

KPI AND HOLDOUT

Rewarded ARPDau and subscription conversion rate among paywall decliners; guardrails: Paid subscription conversion non-inferiority within 3%, D7/D30 user retention, Ad completion rate >= 95%. 20% user-level randomized holdout running for 28 days to measure net revenue impact.

RISKS

- Risk of temporary cannibalization if trial duration is too long
- Potential ad fill variations across international regions

INTEGRATION (@SIMULA/ADS-REACT-NATIVE)

```
import { SimulaAds, useRewardedAd, MiniGameInvitation } from "@simula/ads-react-native";
import { useEffect, useState } from "react";

// P3 on "Janitor Plus Paywall". Trigger: When a free user taps 'Not now' to dismiss the Janitor Plus paywall (moment m3).
// Caps 1/day, 1440 min apart: app remote config (unit cap: dashboard)
const AD_UNIT = "SIM-RWD-janitor-plus-paywall"; // placeholder: create it in the dashboard

export function SubscriptionFeatureSamplingOffer({ userId, partnerImageUrl, grant, onDecline }) {
  const rwd = useRewardedAd(AD_UNIT);
  const [capped, setCapped] = useState(true);
  useEffect(() => {
    SimulaAds.checkFrequencyCap(AD_UNIT, userId).then(setCapped); // true: hide rwd.setMetadata({ surface: "janitor-plus-paywall", proposal: "P3" });
    rwd.load({ charName: "Janitor" }); // Game Partner
  }, [userId]);
  useEffect(() => {
    // Grant on REWARD_VERIFIED (server-verified), never on EARNED_REWARD.
    if (rwd.rewardVerified) grant({ resource: "of1", duration: "1 hour", token: rwd.rewardToken });
  }, [rwd.rewardVerified]);
  // no_fill or not loaded: render nothing; today's path is unchanged.
  if (capped || !rwd.isLoading || rwd.error?.code === "no_fill") return null;
  return (
    <MiniGameInvitation
      isOpen
      titleText={"Try Janitor Plus Free"}
      subText={"Play a quick 15-second game to unlock 1 hour of premium memory."}
      ctaText={"Play Now"}
      charImage={partnerImageUrl}
      onClick={() => rwd.show()}
      onClose={onDecline} // "No Thanks": nothing lost
    />
  );
}
```

WHAT THE JUDGE CHANGED

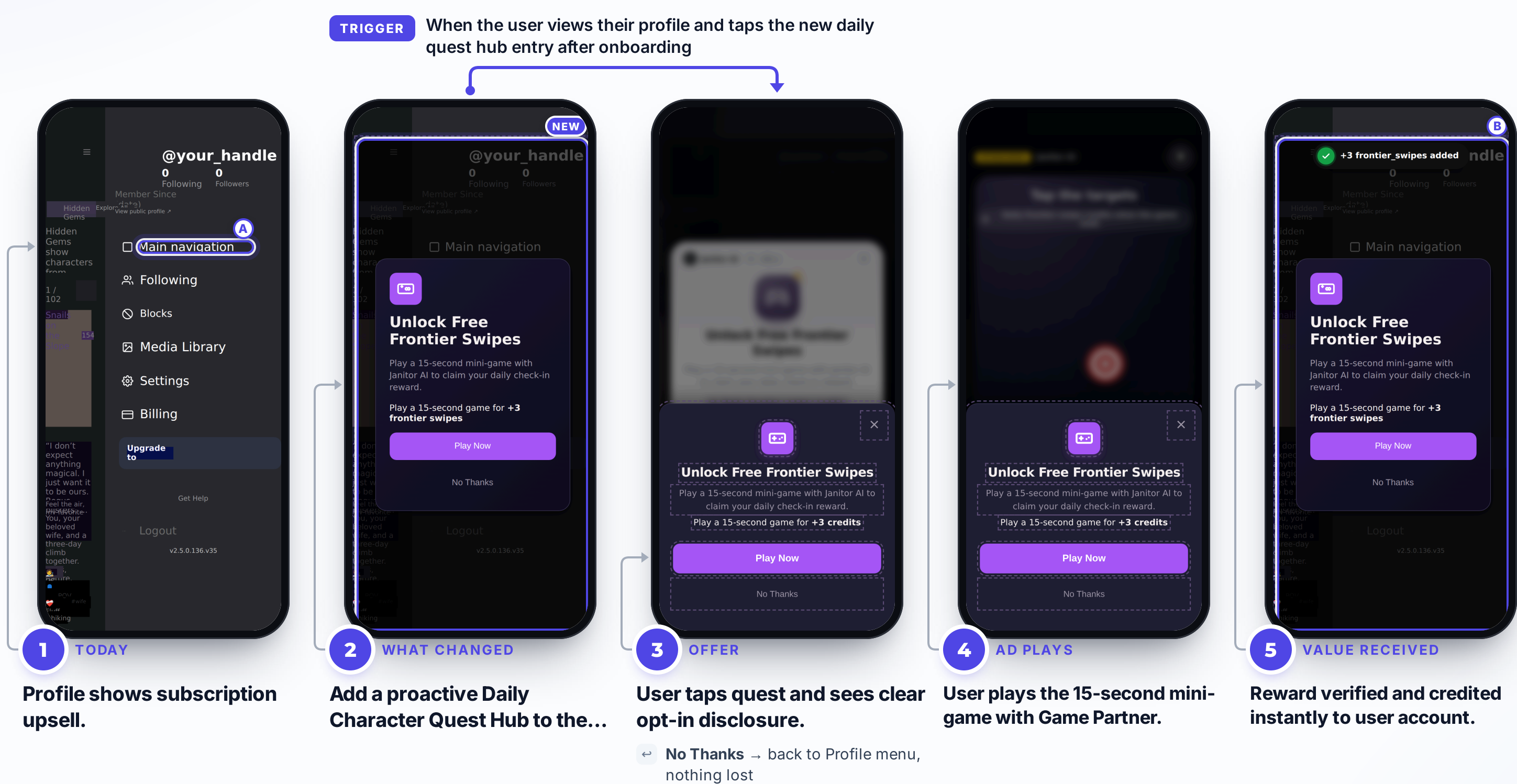
- Round 0 (v1) REVISE 3.70: Cost to serve the reward (\$0.0090) exceeds net revenue per view (\$0.0063) at the low end, making the unit economics... Required: Resize the...
- Round 1 (v2) REVISE 4.60: Economy reference in anchor ('plan') does not match the observed offer identifier ('of1'). Required: Fix the economy reference in the anchor...

PRECEDENTS (KNOWLEDGE BASE)

- **TAX-2** Time-boxed premium unlock or trial ("sponsored session")
- **EX-DUO** Duolingo (education; subscription plus ads plus gems)
- **TRIG-5** Offer UX anatomy (what a good offer card contains)
- **TAX-10** Paywall-decline fallback

Profile menu: play for Daily frontier swipe credits

Earn daily check-in rewards and frontier model swipes by completing character quests via rewarded mini-games.



SHIP Score 4.25 / 5 (ship ≥ 3.8)

Product change

WHY THIS WORKS

\$0.009–\$0.015 per view

Earned per completed US view; the reward costs ≈ \$0.0036 to serve.

Paid path untouched

Restricted to non-payers, capped at 3 daily completions, and sized well below Janitor Plus monthly entitlements.

≤ 3 a day

Opt-in only, at least 30 min apart; nothing plays unless the user taps.

Open the clickable prototype →

Simula SIM-RWD · invitation entry · Game Partner: Janitor AI · min play 15s · reward granted on REWARD_VERIFIED · frame callouts on slide 6

Daily Character Quest Hub

TRIGGER

When the user views their profile and taps the new daily quest hub entry after onboarding.

ELIGIBILITY

Non-paying users.

WHAT THE USER SEES

Unlock Free Frontier Swipes Play a 15-second mini-game with Janitor AI to claim your daily check-in reward.

Play Now

No Thanks

REWARD AND CAPS

Daily frontier swipe credits; granted on REWARD_VERIFIED. At most 3 a day, 30 min apart.

FRAME BY FRAME (SLIDE 5)

- 1

Today Profile shows subscription upsell with no free daily reward option.

A

Upgrade to Janitor Plus only
- 2

What changed Add a proactive Daily Character Quest Hub to the profile.

NEW

Daily Character Quest Hub
- 3

Offer User taps quest and sees clear opt-in disclosure.

Play Now: starts a 15s sponsored game

No Thanks: dismisses the offer

Play 15s to get frontier swipes
- 4

Ad plays User plays the 15-second mini-game with Game Partner.

Mini-game with Janitor AI
- 5

Value received Reward verified and credited instantly to user account.

B

+3 frontier swipes granted

ECONOMICS (COMPUTED IN CODE)

Revenue per completed US view	\$0.009–\$0.015
Reward value at list price	n/a (no priced unit)
Reward ÷ one view	n/a
Max earnable per day (3/day)	n/a
Cheapest pack	none observed
Cost to serve the reward	\$0.0036 per view

SCENARIO, NOT FORECAST

0.375

\$0.0045

\$1,642,500

views per DAU

ARPDau

per year per 1M DAU

Scenario, not a forecast: 25% of DAU engage, 1.5 views each, US mid eCPM after a 25% non-game haircut.

CANNIBALIZATION GUARD

Restricted to non-payers, capped at 3 daily completions, and sized well below Janitor Plus monthly entitlements.

KPI AND HOLDOUT

Rewarded ad ARPDau and daily active engagement rate; guardrails: Paid subscription conversion non-inferiority within -3%, D7 and D30 retention stability. User-level 10% holdout test running for 28 days.

RISKS

- Low risk of subscription cannibalization if daily rewards remain modest compared to Janitor Plus...
- Ad fill rates in non-US regions.

INTEGRATION (@SIMULA/ADS-REACT-NATIVE)

```
import { SimulaAds, useRewardedAd, MiniGameInvitation } from "@simula/ads-react-native";
import { useEffect, useState } from "react";

// P2 on "Profile menu". Trigger: When the user views their profile and taps the new
// daily quest hub entry after onboarding.
// Caps 3/day, 30 min apart: app remote config (unit cap: dashboard)
const AD_UNIT = "SIM-RWD-profile-menu"; // placeholder: create it in the dashboard

export function DailyCharacterQuestOffer({ userId, partnerImageUrl, grant, onDecline }) {
  const rwd = useRewardedAd(AD_UNIT);
  const [capped, setCapped] = useState(true);
  useEffect(() => {
    SimulaAds.checkFrequencyCap(AD_UNIT, userId).then(setCapped); // true: hide
    rwd.setMetadata({ surface: "profile-menu", proposal: "P2" });
    rwd.load({ charName: "Janitor AI" }); // Game Partner
  }, [userId]);
  useEffect(() => {
    // Grant on REWARD_VERIFIED (server-verified), never on EARNED_REWARD.
    if (rwd.rewardVerified) grant({ amount: 3, resource: "frontier_swipes", token: rwd.rewardToken });
  }, [rwd.rewardVerified]);
  // no_fill or not loaded: render nothing; today's path is unchanged.
  if (capped || !rwd.isLoaded || rwd.error?.code === "no_fill") return null;
  return (
    <MiniGameInvitation
      isOpen
      titleText={"Unlock Free Frontier Swipes"}
      subText={"Play a 15-second mini-game with Janitor AI to claim your daily check-in reward."}
      ctaText={"Play Now"}
      charImage={partnerImageUrl}
      onClick={() => rwd.show()}
      onClose={onDecline} // "No Thanks": nothing lost
    />
  );
}
```

WHAT THE JUDGE CHANGED

- Round 0 (v1) REVISE 4.30: Cost to serve the reward (\$0.0090) exceeds net revenue per view (\$0.0063) at the low end, resulting in negative net... Required: Reduce the...

PRECEDENTS (KNOWLEDGE BASE)

- TAX-9** Daily quests, tasks and check-ins (sponsored tasks)
- AI-3**
- EX-DUO** Duolingo (education; subscription plus ads plus gems)
- POL-8** Simula ad units (the client's vocabulary; from public docs and SDK)

Ideas we rejected, and why

Every candidate the judge scored, with its final verdict. Only SHIP becomes a flow; REVISE is never promoted. Weighted score out of 5.

ID	IDEA	VERDICT	SCORE	WHY
P3	Subscription Feature Sampling via Rewarded Ad Product change	SHIP	4.90	SHIP at 4.9 (v3 after 2 revisions). See slide 3.
P2	Daily Character Quest Hub Product change	SHIP	4.25	SHIP at 4.25 (v2 after 1 revision). See slide 5.
P1	Extended Context Unlock Existing mechanic	REJECT	4.80	REJECT after 2 revisions: fixable gate failed: grounding (code); still REVISE after round 2. Top concern: None. The proposal is robust and well-grounded.

Brainstormed, not developed: Paywall-Dcline Session Pass · Frontier Model Swipe Refill · Priority Routing Pass · Sponsored Regeneration Pass · Character Badge Rental

Obvious baseline the proposer had to go beyond: Watch a full-screen interstitial ad before starting any character chat. · Watch an ad to remove all ads across the platform for one day. · Watch an ad to unlock 1 month of unlimited Janitor Plus subscription.

Method and mock fidelity

- 1. **Explore.** 60 steps, 16 states, 60 edges, 0 external surfaces in 5 min; stopped: budget steps. 82 actions deliberately not explored.
- 2. **Understand.** 16 screens in the product model; 3 claims verified against on-screen text, 0 inferred. Synthesized by llm.
- 3. **Mock + QA.** 33% of screens rebuilt in HTML; 45/45 navigation flows pass.
- 4. **Propose.** 8 ideas, 3 developed into typed proposals (patch, storyboard, assumptions). Generated by llm.
- 5. **Judge.** 8 judgment rounds for 3 proposals: code gates first, then a rubric scored by llm; the verdict is computed in code (ship ≥ 3.8 , every criterion ≥ 3 , ≤ 2 rounds).
- 6. **Slides.** Every frame is a screenshot of the patched mock driven through its runtime API; callouts sit on measured DOM positions; numbers come from economics code.

QA FIDELITY (MOCK VS THE REAL SCREENSHOTS)

MEAN COMPOSITE

0.90

FLOW PASS RATE

100%

HTML SCREENS

33%

SCREEN	RENDER	COMPOSITE	ROUNDS	MUST-FIX
Build, Share, Explore	html	0.94	1	0
@your_handle	html	0.92	0	0
More memory for long chats.	html	0.91	1	1
Janitor Plus Paywall	html	0.83	1	1
Explore Characters	image	–	0	0
Search Page	image	–	0	0
Search Characters	image	–	0	0
Search Characters	image	–	0	0
Character Details	image	–	0	0
Character Detail Page	image	–	0	0
Search Characters	image	–	0	0
Search Characters	image	–	0	0

Cost and limitations

MODEL SPEND FOR THIS APP (FROM COST.JSONL)

STAGE	LIVE CALLS	CACHED	TOKENS IN	TOKENS OUT	USD
explore	15	0	49.1k	3.1k	\$0
understand	1	0	20.2k	1.2k	\$0
mock	6	0	35.7k	69.8k	\$0
qa	3	0	13.5k	22.9k	\$0
propose	10	0	532.5k	45.3k	\$0
judge	8	0	425.6k	26.6k	\$0
slides	6	10	48.1k	61.6k	\$0
Total	49	10	1.1M	230.5k	\$0

Gemini free tier: calls cost \$0; tokens are still logged. Models: gemini-3.5-flash-lite, gemini-3.1-flash-lite, gemini-3-flash-preview. Cached calls replay from disk at no cost.

LIMITATIONS

- Explored on an Android emulator (logged-in account); 82 actions left unexplored by guard rails or scope, listed in the model viewer.
- 0 product-model claims inferred rather than observed; the economy review is the human checkpoint for them.
- Revenue per view uses KB eCPM ranges with a non-game haircut; some constants are inferred. ARPDau appears only as a labelled scenario.
- The ad in the frames is simulated by the mock; real fill, creatives and completion rates vary by market.
- The judge is a separate call of the same model family as the proposer; its single-fault calibration table is in judge-eval.md.

How this runs across hundreds of apps

Storage and versioning

- Screenshots, element lists and HTML are content-addressed blobs; artifact JSON references them by hash.
- A model version is keyed by package, version code and a UI fingerprint of the core flows, which also catches over-the-air updates.
- A diff between two model versions (screens, walls, prices, offers) is itself a sales signal.

When to re-explore

- Daily store version poll; weekly no-LLM smoke replay of the core flows and the path to the wall.
- Drift above 10% or any price change triggers an incremental re-explore from the known graph (about 20% of first-run cost).
- A new ad format re-runs only propose and judge.

Where humans review

- App onboarding and login (15 min); economy review, the highest-leverage step (10 min).
- Deck approval before any customer sees it (15–30 min); exception queue for blocked apps.
- Monthly judge audit against labelled outcomes.

Cost and sales fit

- Prospect scan without a mock ≈ \$5 of model calls; full pitch pack ≈ \$20–25 plus 45–60 min of human review.
- The deck, clickable prototype and sizing sheet attach to the CRM deal; each SHIP's integration snippet becomes the implementation ticket.
- Post-launch opt-in, completion and IAP guardrails flow back to recalibrate economics and the judge.